

4D_CCR_1_Ca0Ha0_N1Hn1.azk:

CCR: HN(i)N(i) DD – CA(i-1)HA(i-1) DD
 info: peak H(i)-N(i)-CO(i-1)-CA(i-1): info on psi (i-1)

dimensions: 1: CA (i-1) – CT, do not exceed maxtmax1 evolution time
 2: CO (i-1) – CT/SCT, automatically set according to grid size
 3: N (i) – CT/SCT, automatically set according to grid size

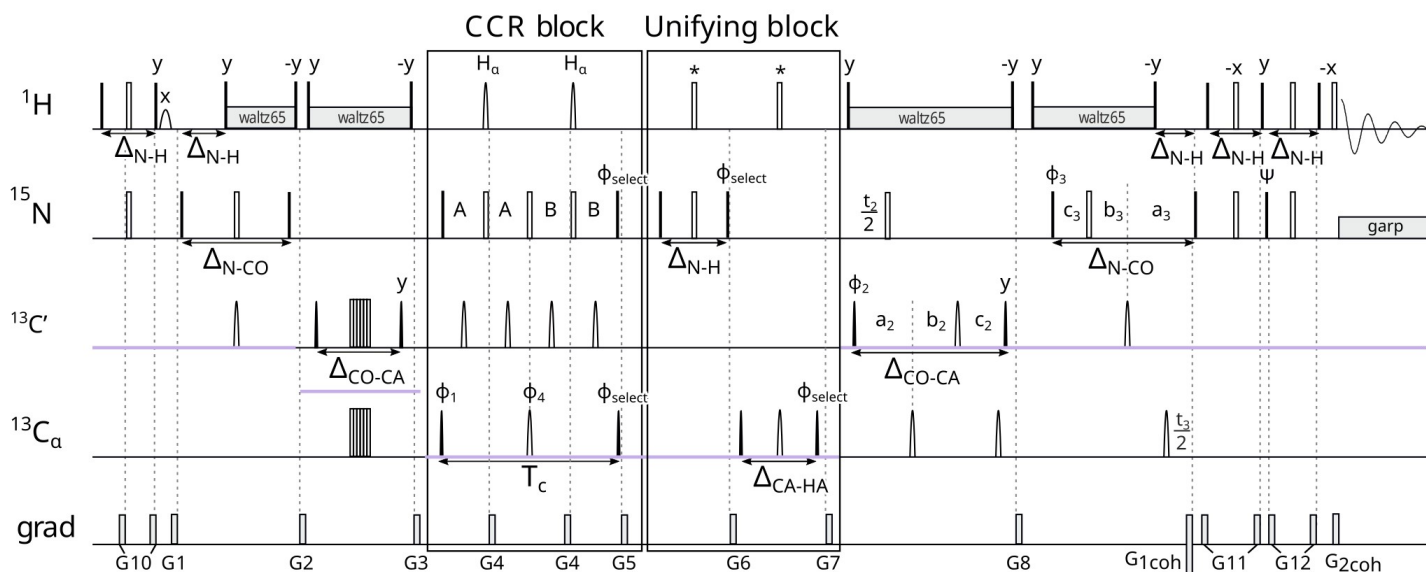
parameters: Tc – CCR evolution time, 0.0286 s – constant, because of CA-CB coupling
 cnst21 – offset CA in ppm [56]
 cnst22 – offset CO in ppm [176]
 cnst19 – offset HA in ppm [4.3]
 cnst18 – bandwidth HA in ppm [3.0]
 cnst18 and cnst19 are specific to a given protein, so they should be set to affect only HA and not HN
 (otherwise, the CCR evolution will be different than we assume)
 DSPFIRM: rectangle
 DIGMOD: baseopt

auto/cross: l0=0 – auto
 l0=1 – cross

set up in TopSpin (after reading the pulse sequence, order important):

getprosol
 getprosol 1H ...u 8W
 wvm -al

$$\text{Gamma} = \frac{+1}{T_c} * \text{arctanh} \frac{I_{\text{trans}}}{I_{\text{ref}}}$$



$\text{H} \rightarrow \text{H}^i\text{N}^i \rightarrow \text{CO}^{i-1}\text{N}^i \rightarrow \text{CA}^{i-1}\text{CO}^{i-1}\text{N}^i \rightarrow \text{CCR} \rightarrow \text{CA}^{i-1}[\text{HA}^{i-1}]\text{CO}^{i-1}\text{N}^i[\text{H}^i] \rightarrow \text{CA}^{i-1}[\text{HA}^{i-1}]\text{CO}^{i-1}\text{N}^i \rightarrow \text{CA}^{i-1}\text{CO}^{i-1}\text{N}^i \rightarrow \text{CO}^{i-1}\text{N}^i \rightarrow \text{H}^i\text{N}^i \rightarrow \text{H}^i$

CCR start coherence: $\text{CA}^{i-1}\text{CO}^{i-1}\text{N}^i$
 minimal CCR start coherence: $\text{CA}^{i-1}\text{N}^i$

Delays:

$$A = (T_c + T_1)/4$$

$$B = (T_c - T_1)/4$$

$$a_2 = \Delta_{CO-CA}/2 + T_2/2$$

$$b_2 = T_2*(1 - \Delta_{CO-CA}/t_{max2})/2$$

$$c_2 = \Delta_{CO-CA}*(1 - T_2/t_{max2})/2$$

$$a_3 = \Delta_{N-CO}/2 + T_3/2$$

$$b_3 = T_3*(1 - \Delta_{N-CO}/t_{max3})/2$$

$$c_3 = \Delta_{N-CO}*(1 - T_3/t_{max3})/2$$

$$\Delta_{N-H} = \Delta_{H-N} = 5.4 \text{ ms}$$

$$\Delta_{N-CO} = 31 \text{ ms}$$

$$\Delta_{CO-CA} = 9.1 \text{ ms}$$

$$\Delta_{CA-HA} = 3.4 \text{ ms} - \text{time for CA} \rightarrow \text{CAHAz and HA} \rightarrow \text{CAzHA evolution not optimal for GLY}$$

Phases:

$$\phi_{\text{select}} = x ('a'), y ('x')$$

$$\phi_1 = x, -x \text{ (ph11 = 0 2)}$$

$$\phi_2 = 2(x), 2(-x) \text{ (ph12 = 0 0 2 2)}$$

$$\phi_3 = x \text{ (ph21 = 0, Phase at start of INEPT before COS-INEPT - for TPPI shifting in T3)}$$

$$\psi = y \text{ (E/A selecting pulse)}$$

$$\phi_4 = 4(x), 4(y), 4(-x), 4(-y) \text{ (ph10 = 0 0 0 0 1 1 1 1 2 2 2 2 3 3 3 3)}$$

$$\text{Receiver: } x, 2(-x), x, -x, 2(x), -x \text{ (ph31 = 0 2 2 0 2 0 0 2)}$$

$$\phi_{\text{receiver}} = \phi_1 + \phi_2 + 2*\phi_4$$

Shapes:

H α - π pulses (Geen and Freeman 1991)

sp26:wvm: reburp(cnst18 ppm, cnst19 ppm)

cnst19: HA offset in ppm [4.3]

cnst18: HA bandwidth in ppm [3.0]

protons decoupling - waltz65

nitrogen decoupling - garp

Q5 - π pulses for CO and CA (Emsley 1969)

Q3 - $\pi/2$ pulses for CO and CA (Emsley 1969)

Evolution:

T1 – constant time evolution of CA during CCR block (States-TPPI)

T2 – constant time or semi-constant time evolution of CO (States-TPPI)

T3 – constant time or semi-constant time evolution of N (Acho-Antiecho)

Tc – evolution of CCR [28.6 ms]

4D_CCR_2_N1Hn1_Ca1Ha1.azk:

CCR: HN(i)N(i) DD – CA(i)HA(i) DD
 info: peak H(i)-N(i)-CO(i-1)-CA(i): info on phi (I)
 CCR start coherence: N(i)CA(i-1)

dimensions: 1: CA (i) – CT, do not exceed maxtmax1 evolution time
 2: CO (i-1) – CT/SCT, automatically set according to grid size
 3: N (i) – CT, do not exceed maxtmax3 evolution time

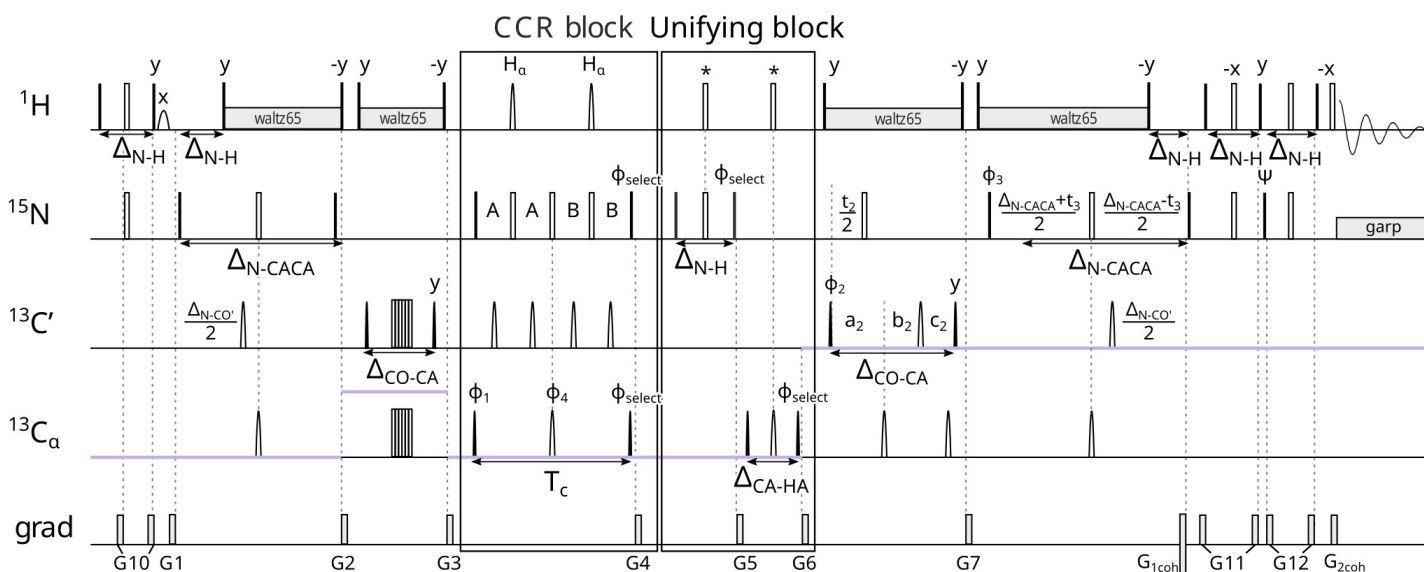
parameters: Tc – CCR evolution time, 0.0286 s – constant, because of CA-CB coupling
 cnst21 – offset CA in ppm [56]
 cnst22 – offset CO in ppm [176]
 cnst19 – offset HA in ppm [4.3]
 cnst18 – bandwidth HA in ppm [3.0]
 cnst18 and cnst19 are specific to a given protein, so they should be set to affect only HA and not HN
 (otherwise, the CCR evolution will be different than we assume)
 DSPFIRM: rectangle
 DIGMOD: baseopt

auto/cross: l0=0 – auto
 l0=1 – cross

set up in TopSpin (after reading the pulse sequence, order important):

getprosol
 getprosol 1H ...u 8W
 wvm -al

$$\text{Gamma} = \frac{+1}{T_c} * \text{arctanh} \frac{I_{\text{trans}}}{I_{\text{ref}}}$$



$H \rightarrow H^i N^i \rightarrow CA^{i-1} CO^{i-1} N^i CA^i \rightarrow CO^{i-1} N^i CA^i \rightarrow \text{CCR} \rightarrow CO^{i-1} N^i [H] CA^i [HA^i] \rightarrow CO^{i-1} N^i CA^i [HA^i] \rightarrow CO^{i-1} N^i CA^i \rightarrow CA^{i-1} CO^{i-1} N^i CA^i \rightarrow H^i N^i \rightarrow H^i$

CCR start coherence: $CO^{i-1} N^i CA^i$
 minimal CCR start coherence: $N^i CA^i$

Delays:

$$A = (T_c + T_1)/4$$

$$B = (T_c - T_1)/4$$

$$a_2 = \Delta_{CO-CA}/2 + T_2/2$$

$$b_2 = T_2^*(1 - \Delta_{CO-CA}/t_{max2})/2$$

$$c_2 = \Delta_{CO-CA}^*(1 - T_2/t_{max2})/2$$

$$a_3 = \Delta_{N-CA}/2 + T_3/2$$

$$b_3 = \Delta_{N-CA}/2 - T_3/2$$

$$\Delta_{N-H} = \Delta_{H-N} = 5.4 \text{ ms}$$

$$\Delta_{N-CACA} = 50 \text{ ms} - \text{time for } N \rightarrow CAzNCAz \text{ evolution (CAi and CAi-1 in one coherence)}$$

$$\Delta_{N-CO}' = 33.5 \text{ ms} - \text{time for } N \rightarrow COzN \text{ evolution when it's not depends of relaxation}$$

$$\Delta_{CO-CA} = 9.1 \text{ ms}$$

$$\Delta_{CA-HA} = 3.4 \text{ ms} - \text{time for } CA \rightarrow CAHAz \text{ and } HA \rightarrow CAzHA \text{ evolution not optimal for GLY}$$

Phases:

$$\phi_{\text{select}} = x \text{ (in 'a'), } y \text{ (in 'x')}$$

$$\phi_1 = x, -x \text{ (ph11 = 0 2)}$$

$$\phi_2 = 2(x), 2(-x) \text{ (ph12 = 0 0 2 2)}$$

$$\phi_3 = x \text{ (ph21 = 0, Phase at start of INEPT before COS-INEPT - for TPPI shifting in T3)}$$

$$\phi_4 = 4(x), 4(y), 4(-x), 4(-y) \text{ (ph10 = 0 0 0 0 1 1 1 1 2 2 2 2 3 3 3 3)}$$

$$\psi = y \text{ (E/A selecting pulse)}$$

$$\text{Receiver: } x, 2(-x), x, -x, 2(x), -x \text{ (ph31 = 0 2 2 0 2 0 0 2)}$$

$$\phi_{\text{receiver}} = \phi_1 + \phi_2 + 2\phi_4$$

Shapes:

H α - π pulses (Geen and Freeman 1991)

sp26:wvm: reburp(cnst18 ppm, cnst19 ppm)

cnst19: HA offset in ppm [4.3]

cnst18: HA bandwidth in ppm [3.0]

protons decoupling - waltz65

nitrogen decoupling - garp

Q5 - π pulses (Emsley 1969).

Q3 - $\pi/2$ pulses (Emsley 1969).

Evolution:

T1 – constant time evolution of CA during CCR block (States-TPPI)

T2 – constant time or semi-constant time evolution of CO (States-TPPI)

T3 – constant time evolution of N (Acho-Antiecho)

Tc – evolution of CCR [28.6 ms]

4D_CCR_3_N0Hn0_N1Hn1.azk:

CCR: HN(i)N(i) DD - HN(i-1)N(i-1) DD
 info: peak H(i)-N(i)-CO(i-1)-CA(i-1): info on phi (i-1) i psi (i-1)

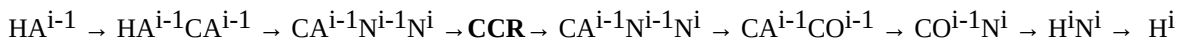
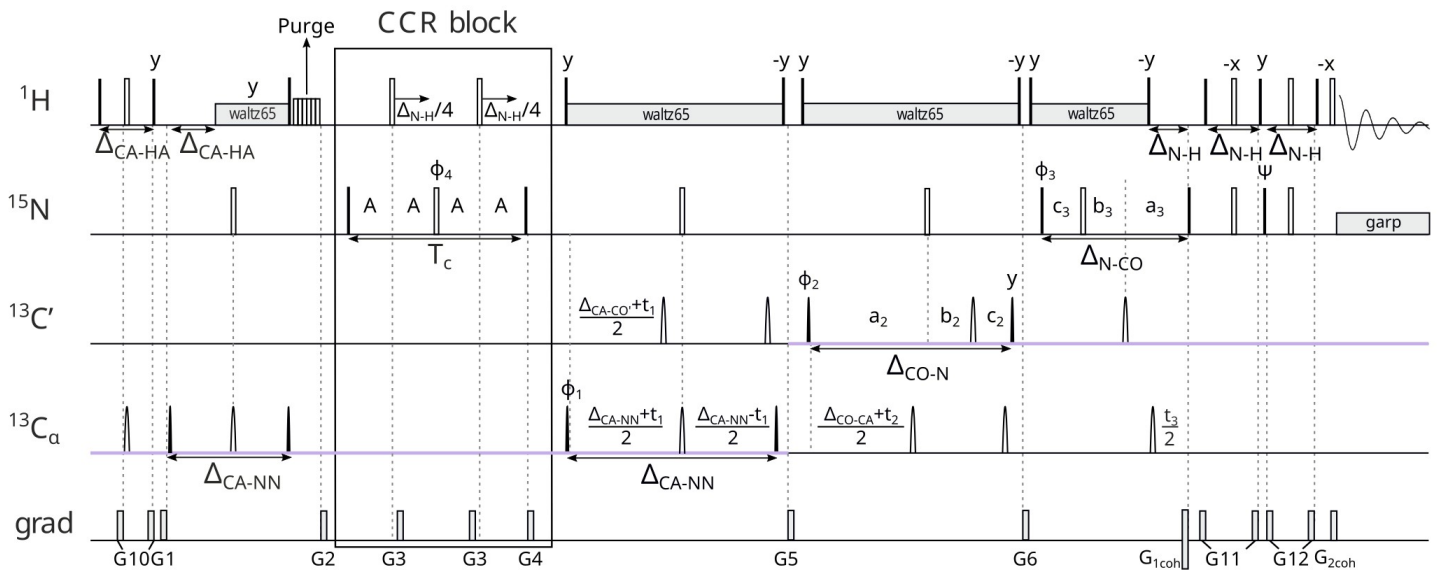
dimensions: 1: CA (i-1) – CT, do not exceed maxtmax1 evolution time – very large, ca. 54 ms
 2: CO (i-1) – CT/SCT, automatically set according to grid size
 3: N (i) – CT/SCT, automatically set according to grid size

parameters: d8 – CCR evolution time, s [0.028]
 cnst21 – offset CA in ppm [56]
 cnst22 – offset CO in ppm [176]
 p17, pl10 – purge
 DSPFIRM: rectangle
 DIGMOD: baseopt

auto/cross: l0=0 – auto
 l0=1 – cross

set up in TopSpin (after reading the pulse sequence, order important):
 (ZGOPTNS -DPURGE) – nie przetestowane
 getprosol
 getprosol 1H ...u 8W
 ZGOPTNS= -DPURGE

$$\text{Gamma} = \frac{+1}{T_c} * \text{arctanh} \frac{I_{trans}}{I_{ref}}$$



CCR start coherence: $\text{CA}^{i-1}\text{N}^{i-1}\text{N}^i$

minimal CCR start coherence: N^{i-1}N^i

Delays:

$$A = T_C/4$$

$$a_1 = \Delta_{N-CA}/2 + T_1/2$$

$$b_1 = \Delta_{N-CA}/2 - T_1/2$$

$$a_2 = \Delta_{CO-N}/2 + T_2/2$$

$$b_2 = T_2*(1 - \Delta_{CO-N}/t_{max2})/2$$

$$c_2 = \Delta_{CO-N}*(1 - T_2/t_{max2})/2$$

$$a_3 = \Delta_{N-CO}/2 + T_3/2$$

$$b_3 = T_3*(1 - \Delta_{N-CO}/t_{max3})/2$$

$$c_3 = \Delta_{N-CO}*(1 - T_3/t_{max3})/2$$

$\Delta_{CA-HA} = 3.4$ ms - time for CA->CAHAz and HA → CAzHA evolution not optimal for GLY

$\Delta_{CA-NN} = 55$ ms – time for CA->NzCANz evolution (Ni and Ni-1 in one coherence)

$\Delta_{CA-CO'} = 9.5$ ms - time for CA->CACOz evolution when it's not depends of relaxation

$\Delta_{CO-CA} = 9.1$ ms

$\Delta_{CO-N} = 29$ ms

$\Delta_{N-CO} = 31$ ms

Phases:

$$\phi_{\text{select}} = x ('a'), y ('x')$$

$$\phi_1 = x, -x \text{ (ph11 = 0 2)}$$

$$\phi_2 = 2(x), 2(-x) \text{ (ph12 = 0 0 2 2)}$$

$$\phi_3 = x \text{ (ph21 = 0, Phase at start of INEPT before COS-INEPT - for TPPI shifting in T3)}$$

$$\psi = y \text{ (E/A selecting pulse)}$$

$$\phi_4 = 4(x), 4(y), 4(-x), 4(-y) \text{ (ph10 = 0 0 0 0 1 1 1 1 2 2 2 2 3 3 3 3)}$$

$$\text{Receiver: } x, 2(-x), x, \text{ (ph31 = 0 2 2 0)}$$

$$\phi_{\text{receiver}} = \phi_1 + \phi_2$$

Shapes:

protons decoupling - waltz65

nitrogen decoupling - garp

Q5 - π pulses (Emsley 1969)

Q3 - $\pi/2$ pulses (Emsley 1969)

Evolution:

T1 – constant time evolution of CA (States-TPPI)

T2 – constant time or semi-constant time evolution of CO (States-TPPI)

T3 – constant time or semi-constant time evolution of N (Acho-Antiecho)

Tc – evolution of CCR [d8, chosen manually]

zgoptns:

-DPURGE: purge water magnetization using B1 and gradient pulses.

4D_CCR_4_Ha0Hn1_CO0.azk:

CCR: HN(i)Ha(i-1) DD – CO(i-1) CSA
 info: peak H(i)-N(i)-CO(i-1)-CA(i-1): info on psi (i-1)

dimensions: 1: CA (i-1) – real time
 2: CO (i-1) – CT, do not exceed maxtmax2 evolution time = 17.6 ms
 3: N (i) - CT/SCT, automatically set according to grid size

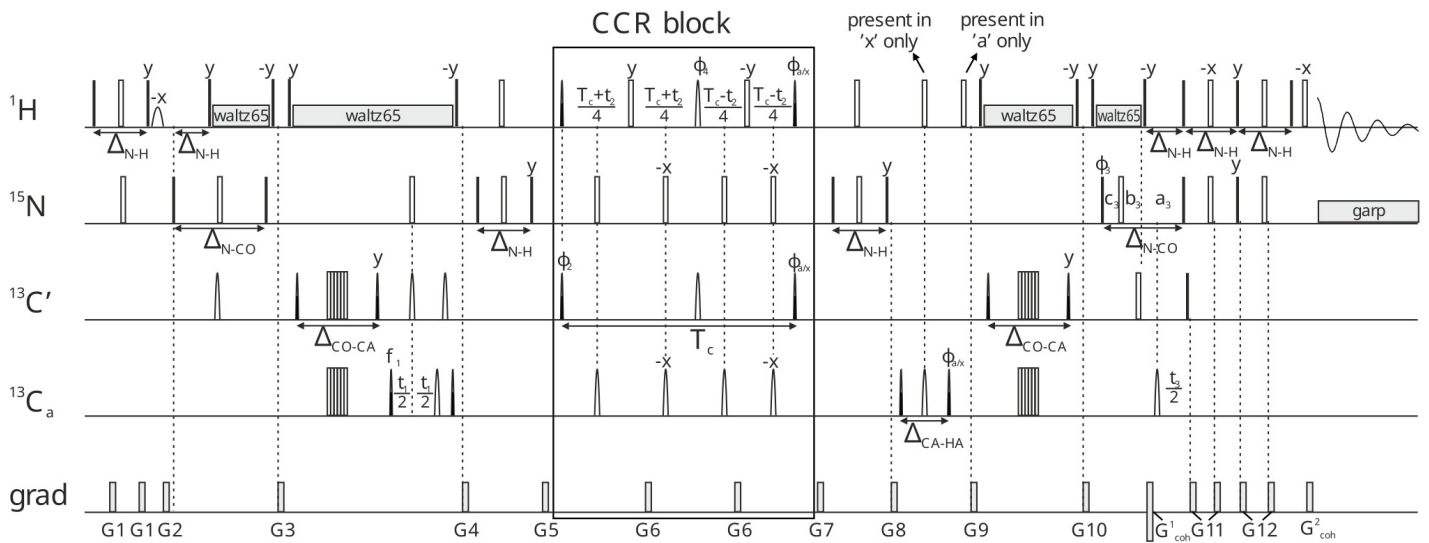
parameters: d8 – CCR evolution time, s [0.028]
 cnst21 – offset CA in ppm [56]
 cnst22 – offset CO in ppm [176]
 cnst19 – offset HN in ppm [8.3]
 cnst18 – bandwidth HN in ppm [3.5]
 cnst18 and cnst19 are specific to a given protein, so they should be set to affect only HN and not HA
 (otherwise, the CCR evolution will be different than we assume)
 DSPFIRM: rectangle
 DIGMOD: baseopt

auto/cross: l0=0 – auto
 l0=1 – cross

set up in TopSpin (after reading the pulse sequence, order important):

getprosol
 getprosol 1H ...u 8W
wvm -al

$$\text{Gamma} = \frac{+1}{T_c} * \text{arctanh} \frac{I_{\text{trans}}}{I_{\text{ref}}}$$



$H \rightarrow H^i N^i \rightarrow CO^{i-1} N^i \rightarrow CA^{i-1} CO^{i-1} N^i \rightarrow CA^{i-1} CO^{i-1} N^i H^i \rightarrow \text{CCR} \rightarrow CA^{i-1} [HA^{i-1}] CO^{i-1} N^i H^i \rightarrow CA^{i-1} [HA^{i-1}] CO^{i-1} N^i$
 $\rightarrow CA^{i-1} CO^{i-1} N^i \rightarrow CO^{i-1} N^i \rightarrow H^i N^i \rightarrow H^i$

CCR start coherence: $CA^{i-1} CO^{i-1} N^i H^i$
 minimal CCR start coherence: $CA^{i-1} CO^{i-1} H^i$

Delays:

$$a_3 = \Delta_{N-CO}/2 + T_3/2$$

$$b_3 = T_3 \cdot (1 - \Delta_{N-CO}/t_{max3})/2$$

$$c_3 = \Delta_{N-CO} \cdot (1 - T_3/t_{max3})/2$$

$$\Delta_{N-H} = \Delta_{H-N} = 5.4 \text{ ms}$$

$$\Delta_{N-CO} = 31 \text{ ms}$$

$$\Delta_{CO-CA} = 9.1 \text{ ms}$$

$$\Delta_{CA-HA} = 3.4 \text{ ms} - \text{time for CA} \rightarrow \text{CAHAz and HA} \rightarrow \text{CAzHA evolution not optimal for GLY}$$

Phases:

$$\phi_{\text{select}} = x ('a'), y ('x')$$

$$\phi_1 = x, -x \text{ (ph11 = 0 2)}$$

$$\phi_2 = 2(x), 2(-x) \text{ (ph12 = 0 0 2 2)}$$

$$\phi_3 = x \text{ (ph21 = 0, Phase at start of INEPT before COS-INEPT - for TPPI shifting in T3)}$$

$$\psi = y \text{ (E/A selecting pulse)}$$

$$\phi_{180} = 4(x), 4(y), 4(-x), 4(-y) \text{ (ph10 = 0 0 0 0 1 1 1 1 2 2 2 2 3 3 3 3)}$$

$$\text{Receiver: } x, 2(-x), x, -x, 2(x), -x \text{ (ph31 = 0 2 2 0 2 0 0 2)}$$

$$\phi_{\text{receiver}} = \phi_1 + \phi_2 + \phi_{180}$$

Shapes:

HN

;sp25:wvm: pc9(cnst18 ppm, cnst19 ppm) – first HN $\pi/2$ pulse in CCR block

;sp26:wvm: reburp(cnst18 ppm, cnst19 ppm) – middle HN π pulse in CCR block

;sp27:wvm: pc9(cnst18 ppm, cnst19 ppm) – last HN $\pi/2$ pulse in CCR block

cnst19: HN offset in ppm [8.3]

cnst18: HN bandwidth in ppm [3.5]

protons decoupling - waltz65

nitrogen decoupling - garp

Q5 - π pulses (Emsley 1969)

Q3 - $\pi/2$ pulses (Emsley 1969)

Evolution:

T1 – real time (States-TPPI)

T2 – constant time evolution of CO during CCR block (States-TPPI)

T3 – constant time or semi-constant time evolution of N (Acho-Antiecho)

Tc – evolution of CCR [d8, chosen manually]

4D_CCR_5_Ca1Ha1_CO1.azk

CCR: CA(i-1)HA(i-1) DD – CO(i-1) CSA
 info: peak H(i)-N(i)-CO(i-1)-CA(i-1): info on psi (i-1)

dimensions: 1: CA (i-1) – CT/SCT, automatically set according to grid size, we recommend not to exceed 10 ms
 2: CO (i-1) – CT/SCT, automatically set according to grid size
 3: N (i) – CT/SCT, automatically set according to grid size

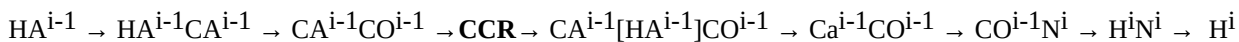
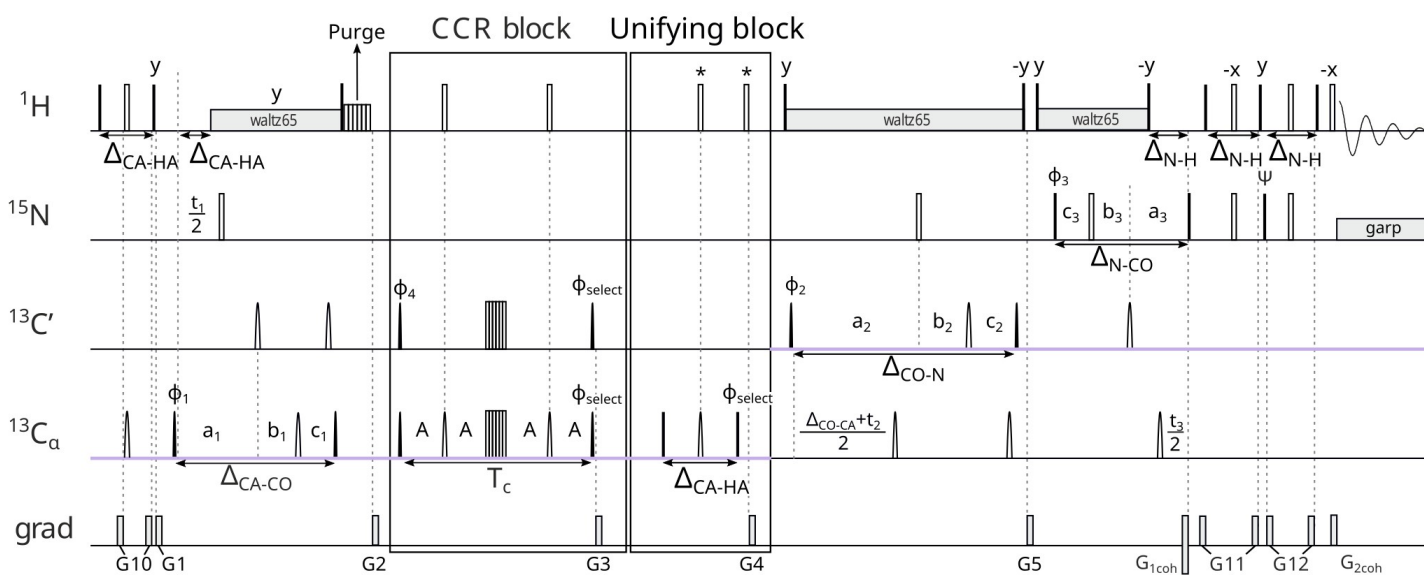
parameters: Tc – CCR evolution time, 0.0286 s – constant, because of CA-CB coupling
 cnst21 – offset CA in ppm [56]
 cnst22 – offset CO in ppm [176]
 pl32, dpres, d1_pres, cnst5 – presaturation
 p17, pl10 – purge
 DSPFIRM: rectangle
 DIGMOD: baseopt

auto/cross: l0=0 – auto
 l0=1 – cross

set up in TopSpin (after reading the pulse sequence, order important):

getprosol
 getprosol 1H ...u 8W
 ZGPTNS= -DPRESAT, -DPURGE

$$\text{Gamma} = \frac{-1}{T_c} * \text{arctanh} \frac{I_{\text{trans}}}{I_{\text{ref}}}$$



CCR start coherence: CAⁱ⁻¹COⁱ⁻¹

minimal CCR start coherence: Caⁱ⁻¹COⁱ⁻¹

Delays:

$$A = T_C/4$$

$$a_1 = T_1/2 + \Delta_{CA-CO}/2$$

$$b_1 = T_1 * (1 - \Delta_{CA-CO}/t_{max1})/2$$

$$c_1 = \Delta_{CA-CO} * (1 - T_1/t_{max1})/2$$

$$a_2 = \Delta_{CO-N}/2 + T_2/2$$

$$b_2 = T_2 * (1 - \Delta_{CO-N}/t_{max2})/2$$

$$c_2 = \Delta_{CO-N} * (1 - T_2/t_{max2})/2$$

$$a_3 = \Delta_{N-CO}/2 + T_3/2$$

$$b_3 = T_3 * (1 - \Delta_{N-CO}/t_{max3})/2$$

$$c_3 = \Delta_{N-CO} * (1 - T_3/t_{max3})/2$$

$$\Delta_{N-H} = \Delta_{H-N} = 5.4 \text{ ms}$$

$$\Delta_{N-CO} = 31 \text{ ms}$$

$$\Delta_{CO-N} = 29 \text{ ms}$$

$$\Delta_{CA-CO} = 6.2 \text{ ms}$$

$$\Delta_{CA-HA} = 3.2 \text{ ms} - \text{time for CA} \rightarrow \text{CAHAz and HA} \rightarrow \text{CAzHA evolution not optimal for GLY}$$

Phases:

$$\phi_{\text{select}} = x ('a'), y ('x')$$

$$\phi_1 = x, -x \text{ (ph11 = 0 2)}$$

$$\phi_2 = 2(x), 2(-x) \text{ (ph12 = 0 0 2 2)}$$

$$\phi_3 = x \text{ (ph21 = 0, Phase at start of INEPT before COS-INEPT - for TPPI shifting in T3)}$$

$$\phi_4 = 4(x), 4(-x) \text{ (ph10 = 0 0 0 0 2 2 2 2)}$$

$$\psi = y \text{ (E/A selecting pulse)}$$

Receiver:

$$\phi_{\text{receiver}} = \phi_1 + \phi_2 + \phi_4 \text{ (ph31 = 0 2 2 0 2 0 0 2)}$$

Shapes:

protons decoupling - waltz65

nitrogen decoupling - garp

Q5 - π pulses (Emsley 1969)

Q3 - $\pi/2$ pulses (Emsley 1969)

Evolution:

T1 – constant time or semi-constant time evolution of CA (States-TPPI)

T2 – constant time or semi-constant time evolution of CO (States-TPPI)

T3 – constant time or semi-constant time evolution of N (Acho-Antiecho)

Tc – evolution of CCR [28.6 ms]

zgoptns:

-DPURGE: purge water magnetization using B1 and gradient pulses.

-DPRESAT: Presaturation of water to reduce the water signal

- d1, cnst5, pl32

4D_CCR_6_CO0_Ca1Ha1.azk

CCR: CA(i)HA(i) DD – CO(i-1) CSA
 info: peak H(i)-N(i)-CO(i-1)-CA(i): info on phi (i)

dimensions: 1: CA (i) – CT/SCT, automatically set according to grid size, we recommend not to exceed constant time (27,2 ms)
 2: CO (i-1) – CT/SCT, automatically set according to grid size
 3: N (i) – CT, do not exceed maxtmax3 evolution time

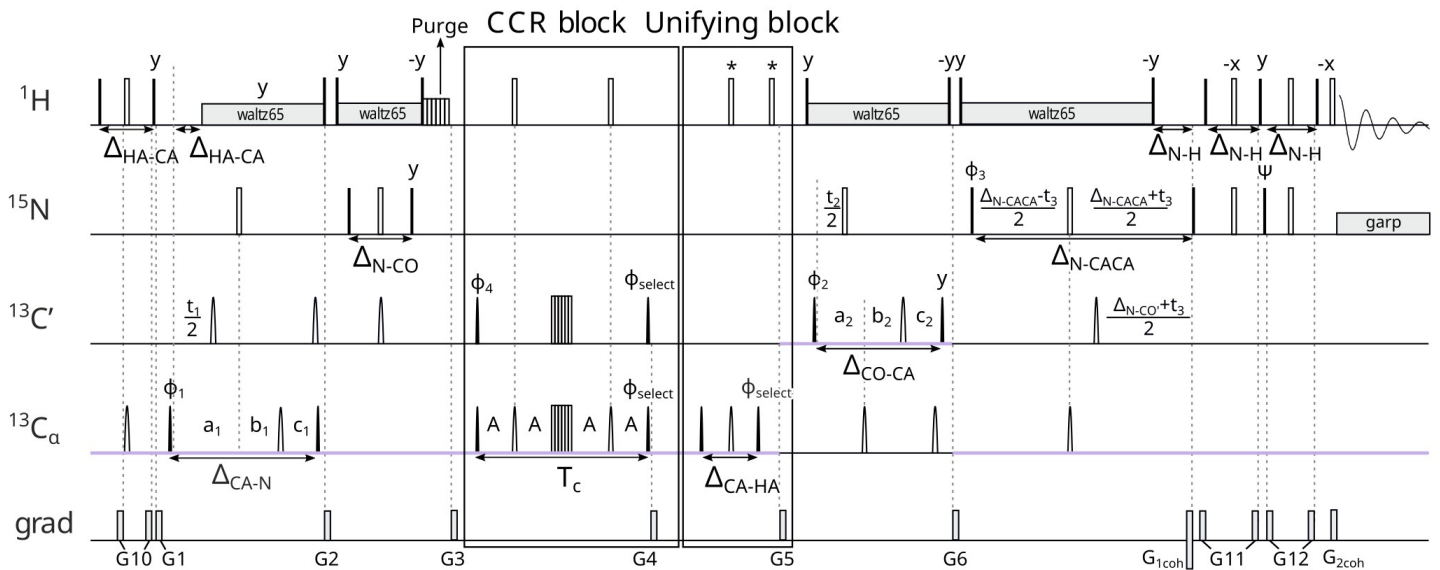
parameters: Tc – CCR evolution time, 0.0286 s – constant, because of CA-CB coupling
 cnst21 – offset CA in ppm [56]
 cnst22 – offset CO in ppm [176]
 pl32, dpres, d1_pres, cnst5 – presaturation
 p17, pl10 – purge
 DSPFIRM: rectangle
 DIGMOD: baseopt

auto/cross: l0=0 – auto
 l0=1 – cross

set up in TopSpin (after reading the pulse sequence, order important):

getprosol
 getprosol 1H ...u 8W
 ZGPTNS= -DPRESAT, -DPURGE

$$\Gamma = \frac{-1}{T_c} * \operatorname{arctanh} \frac{I_{trans}}{I_{ref}}$$



$HA^i \rightarrow HA^iCA^i \rightarrow N^iCA^i + [N^{i+1}CA^i] \rightarrow CO^{i-1}N^iCA^i + [CO^iCA^iN^{i+1}] \rightarrow \mathbf{CCR} \rightarrow CO^{i-1}N^iCA^i[HA^i] + [CO^iCA^i[HA^i]N^{i+1}] \rightarrow CO^{i-1}N^iCA^i + [CO^iCA^iN^{i+1}] \rightarrow CO^{i-1}CA^{i-1}N^iCA^i + [CO^iN^{i+1}] \rightarrow H^iN^i + [CO^iCA^iN^{i+1}CA^{i+1}] \rightarrow H^i + \dots$

CCR start coherence: $CO^{i-1}N^iCA^i + CO^iCA^iN^{i+1}$

minimal CCR start coherence: $CO^{i-1}CA^i$

Delays:

$$A = T_C/4$$

$$a_1 = T_1/2 + \Delta_{CA-N}/2$$

$$b_1 = T_1*(1 - \Delta_{CA-N}/t_{max1})/2$$

$$c_1 = \Delta_{CA-N}*(1 - T_1/t_{max1})/2$$

$$a_2 = \Delta_{CO-CA}/2 + T_2/2$$

$$b_2 = T_2*(1 - \Delta_{CO-CA}/t_{max2})/2$$

$$c_2 = \Delta_{CO-CA}*(1 - T_2/t_{max2})/2$$

$$a_3 = \Delta_{N-CA}/2 + T_2/2$$

$$b_3 = \Delta_{N-CA}/2 - T_2/2$$

$$\Delta_{N-H} = \Delta_{H-N} = 5.4 \text{ ms}$$

$$\Delta_{N-CACA} = 50 \text{ ms} - \text{time for } N \rightarrow CAzNCAz \text{ evolution (CAi and CAi-1 in one coherence)}$$

$$\Delta_{N-CO} = 31 \text{ ms}$$

$$\Delta_{N-CO}' = 33.5 \text{ ms} - \text{time for } N \rightarrow COzN \text{ evolution when it's not depends of relaxation}$$

$$\Delta_{CO-CA} = 9.1 \text{ ms}$$

$$\Delta_{CA-N} = 27.2 \text{ ms}$$

$$\Delta_{CA-HA} = 3.4 \text{ ms} - \text{time for } CA \rightarrow CAHAz \text{ and } HA \rightarrow CAzHA \text{ evolution not optimal for GLY}$$

Phases:

$$\phi_{CCR} = x ('a'), y ('x')$$

$$\phi_1 = x, -x \text{ (ph11 = 0 2)}$$

$$\phi_2 = 2(x), 2(-x) \text{ (ph12 = 0 0 2 2)}$$

$$\phi_3 = x \text{ (ph21 = 0, Phase at start of INEPT before COS-INEPT - for TPPI shifting in T3)}$$

$$\phi_4 = 4(x), 4(-x) \text{ (ph10 = 0 0 0 0 2 2 2 2)}$$

$$\psi = y \text{ (E/A selecting pulse)}$$

Receiver:

$$\phi_{receiver} = \phi_1 + \phi_2 + \phi_4 \text{ (ph31 = 0 2 2 0 2 0 0 2)}$$

Shapes:

protons decoupling - waltz65

nitrogen decoupling - garp

Q5 - π pulses (Emsley 1969)

Q3 - $\pi/2$ pulses (Emsley 1969)

Evolution:

T1 – constant time or semi-constant time evolution of CA (States-TPPI)

T2 – constant time or semi-constant time evolution of CO (States-TPPI)

T3 – constant time evolution of N (Acho-Antiecho)

Tc – evolution of CCR [28.6 ms]

zgoptns:

-DPURGE: purge water magnetization using B1 and gradient pulses.

-DPRESAT: Presaturation of water to reduce the water signal

4D_CCR_7_N1Hn1_CO1.azk

CCR: HN(i)N(i) DD – CO(i) CSA
 info: peak H(i)-N(i)-CO(i)-CA(i): info on phi (i) and psi (i)

dimensions: 1: CA (i) – CT, do not exceed maxmax1 evolution time – very large, ca. 50 ms
 2: CO (i) – CT/SCT, automatically set according to grid size
 3: N (i) – CT/SCT, automatically set according to grid size

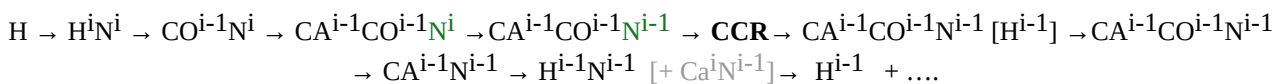
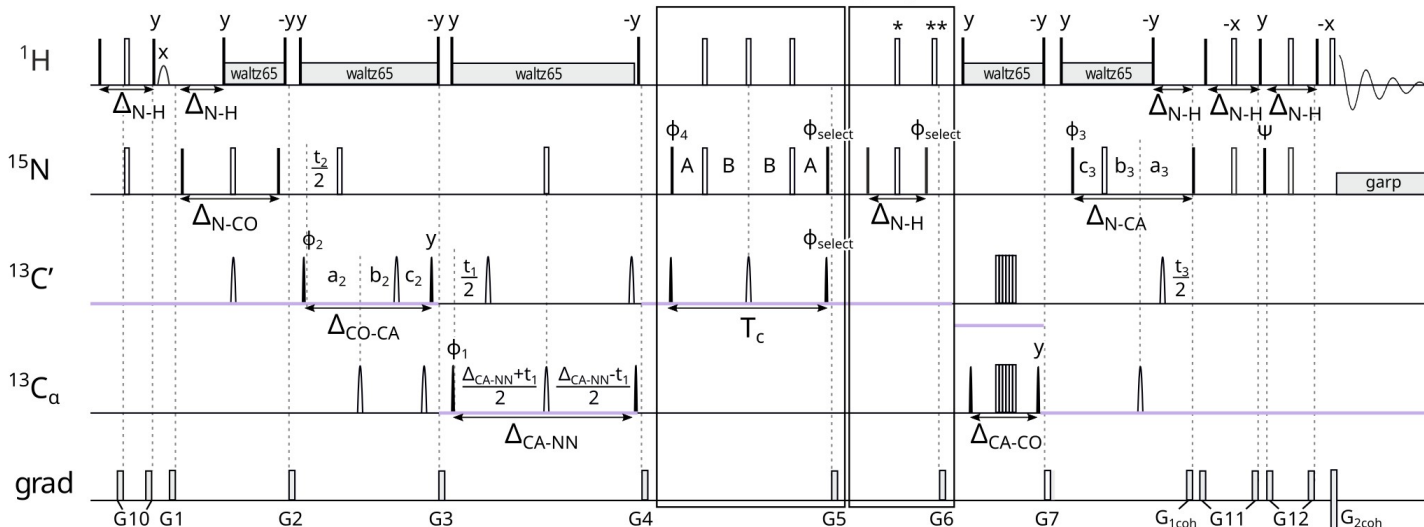
parameters: d8 – CCR evolution time, s [0.028]
 cnst21 – offset CA in ppm [56]
 cnst22 – offset CO in ppm [176]
 DSPFIRM: rectangle
 DIGMOD: baseopt

auto/cross: auto: l0=0
 cross: l0=1

set up in TopSpin (after reading the pulse sequence, order important):
 getprosol
 getprosol 1H ...u 8W

$$\text{Gamma} = \frac{+1}{T_c} * \text{arctanh} \frac{I_{\text{trans}}}{I_{\text{ref}}}$$

CCR block Unifying block



CCR start coherence: $\text{CA}^{i-1}\text{CO}^{i-1}\text{N}^{i-1}$

minimal CCR start coherence: $\text{CO}^{i-1}\text{N}^{i-1}$

Delays:

$$A = (T_c + p14) * 0.25 - p21 / PI$$

$$B = (T_c - p14) * 0.25 + p21 / PI$$

$$a_1 = \Delta_{CA-NN} / 2 + T1 / 2$$

$$b_1 = \Delta_{CA-NN} / 2 - T1 / 2$$

$$a_2 = \Delta_{CO-CA} / 2 + T2 / 2$$

$$b_2 = T2 * (1 - \Delta_{CO-CA} / t_{max2}) / 2$$

$$c_2 = \Delta_{CO-CA} * (1 - T2 / t_{max2}) / 2$$

$$a_3 = \Delta_{N-CA} / 2 + T3 / 2$$

$$b_3 = T3 * (1 - \Delta_{N-CA} / t_{max3}) / 2$$

$$c_3 = \Delta_{N-CA} * (1 - T3 / t_{max3}) / 2$$

$$\Delta_{N-H} = \Delta_{H-N} = 5.4 \text{ ms}$$

$$\Delta_{N-CO} = 31 \text{ ms}$$

$$\Delta_{N-CA} = 28 \text{ ms}$$

$$\Delta_{CO-CA} = 9.1 \text{ ms}$$

$$\Delta_{CA-NN} = 55 \text{ ms} - \text{time for CA} \rightarrow \text{NzCANz evolution (Ni and Ni-1 in one coherence)}$$

$$\Delta_{CA-CO} = 6 \text{ ms}$$

Phases:

$$\phi_{CCR} = x \text{ ('a'), } y \text{ ('x')}$$

$$\phi_1 = x, -x \text{ (ph11 = 0 2)}$$

$$\phi_2 = 2(x), 2(-x) \text{ (ph12 = 0 0 2 2)}$$

$$\phi_3 = x \text{ (ph21 = 0, Phase at start of INEPT before COS-INEPT - for TPPI shifting in T3)}$$

$$\psi = y \text{ (E/A selecting pulse)}$$

$$\phi_4 = x \text{ (ph10 = 0)}$$

$$\text{Receiver: } x, 2(-x), x, \text{ (ph31 = 0 2 2 0)}$$

$$\phi_{\text{receiver}} = \phi_1 + \phi_2$$

Shapes:

protons decoupling - waltz65

nitrogen decoupling - garp

Q5 - π pulses (Emsley 1969)

Q3 - $\pi/2$ pulses (Emsley 1969)

Evolution:

T1 – constant time evolution of CA (States-TPPI)

T2 – constant time or semi-constant time evolution of CO (States-TPPI)

T3 – constant time or semi-constant time evolution of N (Acho-Antiecho)

Tc – evolution of CCR [d8, chosen manually]

4D_CCR_8_CO0_CO1.azk:

CCR: CO(i-1) CSA – CO(i) CSA
 info: peak H(i)-N(i)-CO(i-1)-CA(i): info on phi (i) psi (i)
 CCR start coherence: CO(i-1)N(i)CA(i)CO(i)

dimensions:

- 1: CA (i) – SCT, automatically set according to grid size, we recommend not to exceed 10 ms
- 2: CO (i-1) – SCT, automatically set according to grid size
- 3: N (i) – CT, do not exceed maxtmax3 evolution time

parameters: d8 – CCR evolution time, s [0.028]
 cnst21 – offset CA in ppm [56]
 cnst22 – offset CO in ppm [176]
 DSPFIRM: rectangle
 DIGMOD: baseopt

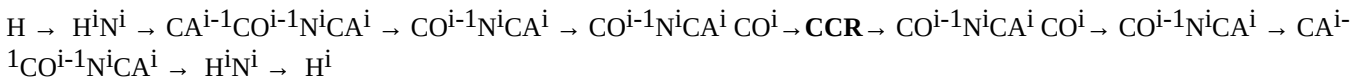
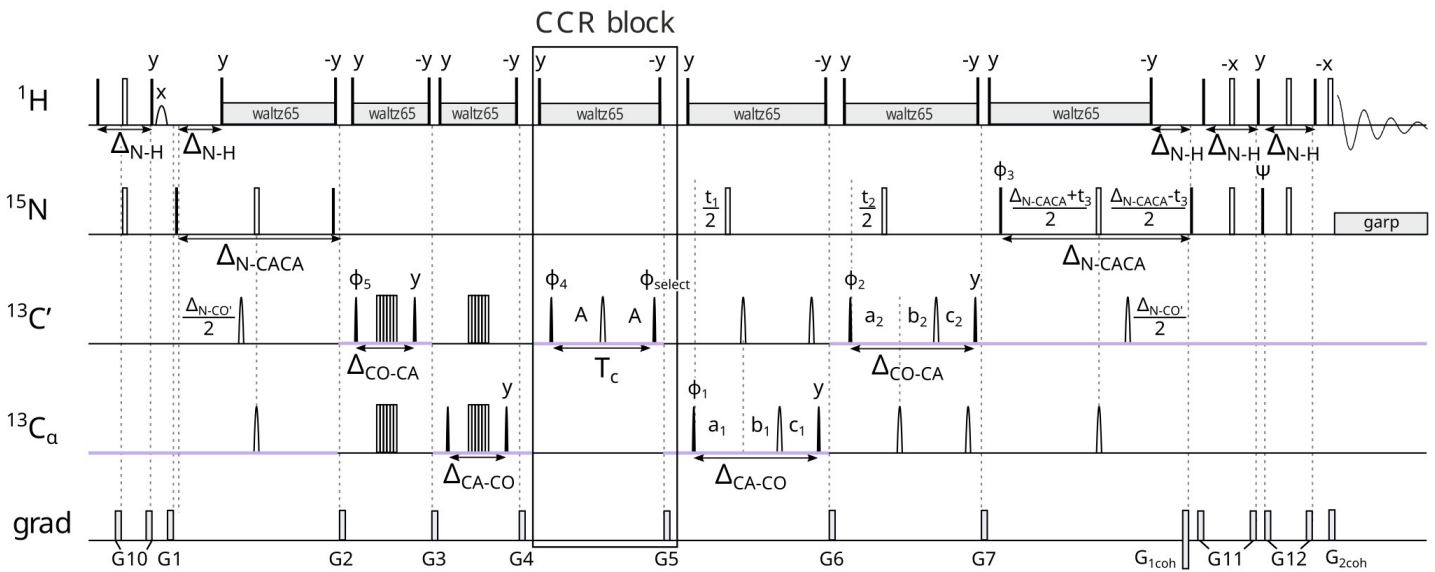
auto/cross: l0=0 – auto
 l0=1 – cross

```

set up in TopSpin (after reading the pulse sequence, order important):
getprosol
getprosol 1H ...u 8W

```

$$Gamma = \frac{+1}{T_c} * \operatorname{arctanh} \frac{I_{trans}}{I_{ref}}$$



CCR start coherence: $\text{CO}^{i-1} \text{N}^i \text{CA}^i \text{CO}^i$
 minimal CCR start coherence: $\text{CO}^{i-1} \text{CO}^i$

Delays:

$$A = T_C/2$$

$$a_1 = \Delta_{CA-CO}/2 + T_1/2$$

$$b_1 = T_1*(1 - \Delta_{CA-CO}/t_{max2})/2$$

$$c_1 = \Delta_{CA-CO}*(1 - T_1/t_{max2})/2$$

$$a_2 = \Delta_{CO-CA}/2 + T_2/2$$

$$b_2 = T_2*(1 - \Delta_{CO-CA}/t_{max2})/2$$

$$c_2 = \Delta_{CO-CA}*(1 - T_2/t_{max2})/2$$

$$a_3 = \Delta_{N-CA}/2 + T_3/2$$

$$b_3 = \Delta_{N-CA}/2 - T_3/2$$

$$\Delta_{N-H} = \Delta_{H-N} = 5.4 \text{ ms}$$

$$\Delta_{N-CACA} = 50 \text{ ms} - \text{time for } N \rightarrow CAzNCAz \text{ evolution (CAi and CAi-1 in one coherence)}$$

$$\Delta_{N-CO}' = 33.5 \text{ ms} - \text{time for } N \rightarrow COzN \text{ evolution when it's not depends of relaxation}$$

$$\Delta_{CO-CA} = 9.1 \text{ ms}$$

$$\Delta_{CA-CO} = 6 \text{ ms}$$

Phases:

$$\phi_{CCR} = x \text{ (in 'a'), } y \text{ (in 'x')}$$

$$\phi_1 = x, -x \text{ (ph11 = 0 2)}$$

$$\phi_2 = 2(x), 2(-x) \text{ (ph12 = 0 0 2 2)}$$

$$\phi_3 = x \text{ (ph21 = 0, Phase at start of INEPT before COS-INEPT - for TPPI shifting in T3)}$$

$$\phi_5 = 2(x), 2(-x) \text{ (ph16 = 0 0 0 0 0 0 0 2 2 2 2 2 2 2)}$$

$$\phi_4 = 2(x), 2(-x) \text{ (ph15 = 0 0 0 0 0 0 0 0 0 0 0 0 2 2 2 2 2 2 2 2 2 2 2 2)}$$

$$\psi = y \text{ (E/A selecting pulse)}$$

$$\phi_4 = 4(x), 4(y), 4(-x), 4(-y) \text{ (ph10 = 0 0 0 0 1 1 1 1 2 2 2 2 3 3 3 3)}$$

$$\text{Receiver: } x, 2(-x), x, -x, 2(x), -x, -x, 2(x), -x, x, 2(-x), x \text{ (ph31 = 0 2 2 0 2 0 0 2 2 0 0 2 0 2 2 0)}$$

$$\phi_{\text{receiver}} = \phi_1 + \phi_2 + \phi_5$$

Shapes:

protons decoupling - waltz65

nitrogen decoupling - garp

Q5 - π pulses (Emsley 1969).

Q3 - $\pi/2$ pulses (Emsley 1969).

Evolution:

T1 – semi-constant time evolution of CA (States-TPPI)

T2 – constant time or semi-constant time evolution of CO (States-TPPI)

T3 – constant time evolution of N (Acho-Antiecho)

Tc – evolution of CCR [28 ms]